QP CODE: 25900061

MGU-UGP (HONOURS) REGULAR EXAMINATIONS MARCH 2025

Second Semester
DSC – MG2DSCECT101 – DATA COMMUNICATION
2024 Admission Onwards
Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest(I) and Appreciation (Ap)

Students should Attempt at least one question from each course outcome to enhance their overall outcome attainability.

SCHEME OF EVALUATION

Part A

Multiple Choice Questions
Answer all questions.Each questions carries 1 Mark

		Learning Domain	CO No(s)
1.	C = B * log2(1 + S/N)	[K]	CO 1
2.	Wavelength decreases	[U]	CO 1
3.	Simplex communication	[U]	CO 1
4.	Focus signals into narrow beams	[U]	CO 2
5.	Bayonet locking mechanism	[U]	CO 2
6.	Both Amplitude and phase	[K]	CO 3
7.	USB (Universal Serial Bus)	[U]	CO 3
8.	Data is transmitted in parallel over multiple lines	[U]	CO 3
9.	DSSS	[U]	CO 3
10.	Baud rate represents the number of signal units per second, while bit rate represents the number of bits per second	[U]	CO 3

[1x10 = 10]

Part B

Short Answer Questions
Answer any 4 questions.
Each questions carries 5 Marks

11. Describe the relationship between bitrate and bandwidth in a communication system. [U] CO 1

Bit rate:
This refers to the number of bits (0s and 1s) transmitted per unit of time, typically measured in bits per second (bps). It directly reflects the amount of data being sent.
Baud rate:
This refers to the number of symbols (or signal changes) transmitted per unit of time, typically measured in baud or symbols per second. A symbol can represent one or more bits.
Explanation – 3 Marks

The relationship between bit rate and baud rate
Bit rate = Baud rate * Number of bits per symbol
Explanation – 3 Marks
Relationship – 2 Marks

12. Explain the different types of data representation in communication systems with examples. [U] CO 1

Data representations are in different forms such as text, numbers, images, audio, and video.

Explanation with example 1 mark for each.

13. How to reduce the signal loss in coaxial cable? [U] CO 2

To reduce signal loss in coaxial cable, use a higher quality cable with lower resistance conductors and shielding, and minimize cable length and the number of connectors.
Explanation – 5 Marks – Value liberally

14. Describe the working of Adaptive Delta Modulation (ADM). How does it differ from standard Delta Modulation (DM)? [U] CO 3

Working of Analog to Digital Modulation – 4 Marks
Difference – 1 Mark

15. Explain the need for Analog to Digital Conversion in communication systems. [U] CO 3

Explanation for ADC (Pulse Code Modulation and Delta Modulation) – 4 Marks
Need – 1 Mark

16. Explain the concept of multiplexing and its types. [U] CO 3

Explanation for Multiplexing – 1 Mark
FDM, WDM, Synchronous TDM, Statistical TDM – 1 Mark for each

[5x4= 20]

Part C

Essay Questions
Answer any 2 questions.
Each questions carries 10 Marks

17. Discuss the primary types of transmission impairments in communication system and explain their causes, effects on signal integrity, and common mitigation techniques. [U] CO 1

Types of transmission impairments
Attenuation, Distortion and Noise – 2.5 Marks for each
Causes – 2.5 Marks

18. Describe the construction of a twisted pair cable, including the purpose of twisted the wires and the different types of shielding (UTP and STP). [U] CO 2

Explanation of Twisted Pair Cable– 4 marks
Explanation of UTP and STP – 6 Marks

19. Discuss the advantage, disadvantage and applications of ASK and FSK. [U] CO 3

Amplitude Shift Keying -5 Marks

(Working, Applications, Advantages and Disadvantages)

Frequency Shift Keying - 5 Marks

(Working, Applications, Advantages and Disadvantages)

[10x2 = 20]